

Software Development Prompt Worklog

Databricks Student Attrition Risk Application

Feature-003 - Structured Advisor Briefing Workflow Testing / US-18

Development method: Claude Code AI agent + GitHub Spec Kit / specification-driven development

Human reviewer: Renny

Purpose: A concise but evidentiary record of the prompts, human decisions, generated development artifacts, testing outputs, and manual verification used for Feature-003.

Human review declaration: This worklog records the prompts, AI-generated artifacts, design decisions, analysis findings, implementation guidance, and manual verification used for Feature-003. Human decisions and the final verification findings were recorded by Renny.

1. How to Read This Worklog

This document deliberately separates prompt evidence from output summaries.

- The main worklog condenses what happened at each Feature-003 development stage.
- Each prompt-driven stage identifies the prompt ID that drove the work.
- Human Review records the decisions and review points supplied by Renny.
- Appendix A preserves the full prompt text used in the workflow so the document remains a genuine prompt worklog rather than only a retrospective description.
- Appendix B preserves the manual verification guide used after implementation.
- Appendix C reproduces the human verification findings verbatim, except for the requested username replacement.

Development sequence

```
Repository Investigation
-> Human Scope Decisions
-> Specification
-> Clarification
-> Implementation Plan
-> Tasks
-> Analysis and Remediation
-> Implementation
-> Manual Verification
-> Human Findings
```

2. Feature-003 - Structured Advisor Briefing Workflow Testing / US-18

2.1 Repository Investigation

Prompt Used

P01 - Repository Investigation

Full prompt: Appendix A - P01

Output

The repository and relevant project documentation were inspected without modifying files. The review focused on architecture/components, dependencies, orchestration/control flow, interfaces/contracts, and testing boundaries for US-18.

- The real Generative AI briefing implementation was not yet integrated.
- Final Structured Advisor Briefing acceptance criteria were not yet available.
- Controlled briefing-generation and validation outcomes were therefore identified as the approved way to test workflow behaviour.
- Six human decisions (H1-H6) were raised to settle scope, defect handling, reuse boundaries, UI exclusion, legacy success criteria, and concurrency scope.

Human Review

Repository findings were reviewed and the H1-H6 scope decisions were supplied by Renny before specification work continued.

2.2 Human Scope and Behavioural Decisions

The six repository-investigation decisions established the Feature-003 boundary:

- H1 - Feature-003 covers the briefing workflow only; health, high-risk list and student profile are out of scope.
- H2 - The blank/whitespace briefing mismatch was initially to be represented as an expected-fail test and raised as a defect.
- H3 - Only new Feature-003 files may be added; existing Feature-001 and Feature-002 test files remain unchanged and already-covered behaviour is not duplicated.
- H4 - ui.py / Streamlit testing is excluded because advisor-facing dashboard testing belongs to US-17.
- H5 - Feature-001 SC-006 is closed by Feature-003; SC-007 is excluded with its reason recorded.
- H6 - Concurrent same-student requests are excluded.

Human Review

Scope and behavioural decisions authored/reviewed by Renny.

2.3 Feature-003 Specification

Prompt Used

P02 - Feature-003 Specification (/speckit-specify)

Full prompt: Appendix A - P02

Output

The specification defined Feature-003 as a testing-only story for the briefing generation, validation, retry, error-handling and validated-storage workflow across the approved service, REST and MCP boundaries.

It required controlled test doubles, scenario-driven workflow coverage, call-order verification, retry-feedback propagation, storage decisions, log hygiene, store parity, traceability, SC-006 verification, and explicit deferral of later backlog work.

The initial specification raised two final clarification points: confirmation of the exact Product Backlog US-18 boundary, and the amount of cross-boundary re-verification required.

Human Review

Renny supplied the exact US-18 story and acceptance criterion and confirmed that the four Feature-003 user stories matched the intended scope, while existing cross-boundary evidence should be tracked rather than unnecessarily duplicated.

2.4 Feature-003 Clarification

Prompt Used

P03 - Feature-003 Clarification (/speckit-clarify)

Full prompt: Appendix A - P03

Output

Three remaining decisions were resolved and integrated into the specification:

- C1 - Defects are recorded both through the test annotation/verification and the traceability record.
- C2 - The traceability record may be hand-maintained, but an automated existence check must confirm every cited scenario still resolves.
- C3 - Governed-store parity re-runs the six approved storage-decision outcomes rather than every Feature-003 scenario.

Human Review

The recommended options for C1-C3 were selected and approved by Renny, prioritising simple solutions and maintaining traceability.

2.5 Feature-003 Implementation Plan

Prompt Used

P04 - Feature-003 Implementation Plan

Full prompt: Appendix A - P04

Output

The plan preserved Feature-003 as a testing-only implementation. Production source files and existing Feature-001/Feature-002 test files remained read-only.

- Reuse the existing application architecture and test conventions.
- Use controlled generation and validation doubles instead of unfinished US-13/US-14 implementations.
- Plan verification for call ordering, retry feedback, storage write counts, valid-only persistence, log hygiene, store parity, SC-006, defect lifecycle, and scenario-to-requirement traceability.
- Keep US-12, US-13, US-14, US-16, Streamlit/UI, concurrency, SC-007 and production defect fixes out of scope.

Human Review

The implementation plan and its scope restrictions were reviewed as the basis for task generation.

2.6 Feature-003 Tasks

Prompt Used

P05 - Feature-003 Tasks

Full prompt: Appendix A - P05

Output

tasks.md translated the approved specification and plan into dependency-ordered implementation work without expanding Feature-003 scope.

The later consistency analysis confirmed 34 tasks, 41 functional requirements, 16 success criteria, 100% requirement coverage, and no unmapped tasks.

Human Review

Task scope retained the testing-only boundary: new Feature-003 test/support and traceability files only, with production source and existing Feature-001/002 tests read-only.

2.7 Pre-Implementation Analysis and Remediation

Prompt Used

P06 - Pre-Implementation Analysis and Remediation

Full prompt: Appendix A - P06

Output

The analysis found no critical issues, no constitution violations, no scope leakage, no duplicated tasks, and 100% requirement coverage.

- F1 - High: task dependency ordering around traceability.md needed correction before concurrent streams could append to it.
- F2 - Medium: FR-013-FR-020 needed explicit traceability identifiers rather than only a range reference.
- F3 - Medium: success-criterion identifiers needed to be added to the tasks that already implemented them.
- F4 - Medium: one minor specification wording issue was already resolved by FR-005 and did not block implementation.
- F6/F7 - Low: Phase 6 placement and the absence of [P] markers were accepted as dependency-correct and required no change.

The analysis also confirmed that tracked re-verification was supported by the specification and that controlled test doubles were not being misrepresented as real US-13/US-14 implementations.

Human Review

Analysis findings were recorded before implementation so task dependency and traceability corrections could be applied without changing Feature-003 behaviour.

2.8 Blank-Content Defect Resolution Before Feature-003 Implementation

Before Feature-003 implementation, the previously recorded blank/whitespace briefing defect was resolved separately against the Feature-002 specification. The latest Feature-003 artifacts therefore superseded the earlier expected-fail approach: the scenario became an ordinary passing verification, while the defect remained recorded in traceability as resolved.

Scope Effect

- Feature-003 remained testing-only.
- The production fix was not reimplemented as part of Feature-003.
- The resolved defect continued to be retained as audit evidence rather than erased.

2.9 Feature-003 Implementation

Prompt Used

P07 - Feature-003 Implementation

Full prompt: Appendix A - P07

Output

The implementation prompt constrained work to the approved Feature-003 test/support and traceability files, required controlled outcomes, preserved production and existing Feature-001/002 tests as read-only, and required all planned pytest/Ruff validation gates.

The implementation stage also required a human-review overview and a practical manual verification guide so the completed testing feature could be independently checked.

Human Review

The source worklog records a subsequent 12-step manual verification process performed by Renny.

2.10 Manual Verification

The source worklog records the following post-implementation checks and results:

- Full suite: 143 tests passed.
- Feature-003 test subset: 59 tests passed.
- Scoped Git checks confirmed no modified production source files were identified by the verification commands.
- Successful workflow ordering test passed.
- Retry-feedback propagation test passed.
- The 11-outcome workflow matrix was present and passed.
- Storage-decision verification returned 10 passes.
- Governed-store parity returned 6 passes.
- Boundary/log-hygiene verification returned 7 passes; REST re-verification was intentionally omitted and tracked in traceability.md.
- The traceability self-check was deliberately broken to produce one failure, then restored to one pass.
- The full 143-test suite passed again without network connectivity.
- traceability.md was reviewed to distinguish completed, tracked-re-verification and deferred behaviour.

3. Current Feature-003 Status

Area	Status
Feature	Feature-003 / US-18 - Structured Advisor Briefing Workflow Testing
Feature type	Testing-only / test-architecture feature
Functional requirements	41
Success criteria	16
Generated tasks	34
Requirement coverage	100% according to pre-implementation analysis
Feature-003 automated subset	59 passed
Full automated suite	143 passed
Offline full-suite verification	143 passed
Governed-store parity	6 approved outcomes passed
Blank/whitespace defect	Resolved separately before Feature-003 implementation; verified as ordinary passing behaviour
Production source / prior tests	Read-only within Feature-003 scope
Deferred	US-12 prompt content; US-13 real AI; US-14 semantic validation; US-16 live end-to-end integration; Streamlit/UI; concurrency; SC-007

4. Human Review Record

The Feature-003 process intentionally retained human control over scope, clarification, verification and acceptance of AI-generated work.

Human reviewer: Renny

- repository findings and H1-H6 scope decisions;
- Product Backlog US-18 story and acceptance criterion confirmation;
- C1-C3 clarification decisions;
- specification, planning and task-scope review;
- pre-implementation analysis findings;
- manual verification of the implemented Feature-003 test suite;
- review of traceability.md completed, tracked-re-verification and deferred sections.

The AI agent was explicitly instructed not to perform commits, pushes, merges, rebases, branch switches, pull requests or other Git/GitHub write operations during the documented workflow.

Appendix A - Full Prompt Log

The following appendix preserves the actual prompt text used in the Feature-003 workflow. Outputs are summarised in the main body to keep the worklog readable.

P01 - Repository investigation for Feature-003

Scope: Feature-003

Human review: Reviewed by Renny as part of the documented Feature-003 workflow.

Prompt Used (verbatim)

Inspect the repository and relevant project documentation to prepare for Feature-003: US-18: Structured Advisor Briefing Workflow Testing specification development.

Do not modify any files.

Inspect the existing application architecture, Feature-001 and Feature-002 implementation, existing tests, and relevant project documentation.

Determine what can already be established about:

architecture/components involved in briefing generation, validation, retry, error handling, and storage;

dependencies required by US-18, including which dependencies already exist and which are currently unavailable;

orchestration/control flow for successful briefing generation, validation failure, retry, terminal failure, and validated storage;

interfaces/contracts between generation, validation, retry workflow, persistence, and application orchestration;

testing boundaries/mocking, including what can be tested using controlled test doubles and what must wait for later integration.

Account for the following current constraints:

the real Generative AI briefing implementation is not yet integrated;

final Structured Advisor Briefing acceptance criteria are not yet available;

real semantic briefing validation therefore cannot currently be tested;

US-18 workflow testing may use controlled briefing-generation and validation outcomes to test successful and failed workflow behaviour.

For every relevant finding, classify it as:

CONFIRMED FROM REPOSITORY
Directly supported by existing code, tests, or project documentation.

REUSABLE EXISTING CAPABILITY
Existing architecture or functionality that Feature-003 can use without reimplementing it.

FEATURE-003 IMPLEMENTATION NEED
Functionality or testing capability that would need to be added specifically for US-18.

DEFERRED DEPENDENCY
Functionality that cannot currently be exercised because it depends on the real Generative AI integration, final briefing acceptance criteria, or another unfinished user story.

UNRESOLVED PRODUCT DECISION
A behavioural or acceptance decision that cannot be determined from the repository or approved documentation and requires my decision.

Explain how US-18 fits into the existing architecture and how the complete testable workflow would operate.

Identify what parts of US-18 can be completed now and what parts must be deferred.

Briefly explain how the findings should later inform:

/speckit-specify

/speckit-clarify

/speckit-plan

/speckit-tasks

Do not create Feature-003 yet.

Do not modify the constitution, specification, plan, tasks, tests, or implementation files.

Do not propose unnecessary architectural changes or duplicate functionality already implemented by Feature-001 or Feature-002.

Do not perform Git operations.

P02 - Generate Feature-003 specification

Scope: Feature-003

Human review: Reviewed by Renny as part of the documented Feature-003 workflow.

Prompt Used (verbatim)

/speckit-specify

Create the Feature-003 specification for US-18: Structured Advisor Briefing Workflow Testing using:

the project constitution v1.1.0;

the approved Product Backlog US-18 user story;

the confirmed Feature-003 repository investigation and scope decisions;

merged Feature-001 (US-08) and Feature-002 (US-15) artifacts and source as implementation context only.

Treat the approved Feature-003 scope decisions as authoritative. Do not reopen them unless they directly conflict with the constitution or Product Backlog.

Feature-003 is a test-architecture story only. It adds no production component. The briefing generation, validation, retry, error-handling and validated-storage architecture already exists through Feature-001 and Feature-002.

Define WHAT Feature-003 must verify, including its observable testing behaviour, acceptance criteria, boundaries, dependencies, failure behaviour and traceability.

Feature-003 scope is limited to the Structured Advisor Briefing workflow across the service, REST and MCP boundaries. Non-briefing functionality such as health, high-risk-list and student-profile behaviour is out of scope.

Feature-003 must be purely additive:

do not modify production source files;

do not modify Feature-001 or Feature-002 test files;

do not duplicate behaviour already adequately verified by those suites;

additional Feature-003 coverage must use new test files while reusing existing testing conventions and doubles where appropriate.

Generation and validation outcomes must use controlled test doubles. This is the approved testing approach because concrete US-13 Generative AI generation and US-14 semantic validation are not yet implemented. Do not invent Structured Advisor Briefing acceptance criteria. Any acceptance-criteria values required by test fixtures must be visibly synthetic, such as `PLACEHOLDER_CRITERION_A`.

The specification must cover the complete testable briefing workflow, including:

successful first-attempt generation and validated storage;

validation failure followed by successful retry;

terminal generation failure;

terminal validation failure;

configuration failure;

storage failure;

not-at-risk refusal;

unknown student behaviour;

get-or-create behaviour for an existing validated briefing;

explicit regeneration.

Where applicable, these behaviours must be demonstrated consistently across the service, REST and MCP boundaries.

Feature-003 must also verify:

workflow/seam call ordering rather than only invocation counts;

propagation of actual validation feedback into the retry generation request;

storage decisions and expected write counts;

failed or invalid outputs are not stored as validated briefings;

previously stored valid briefings are preserved where required;

error mapping;

metadata-only log hygiene;

behavioural parity between in-memory and Volume-backed stores using controlled test infrastructure;

traceability from test scenarios to requirements, success criteria and relevant backlog stories.

Include the following existing traceability gaps:

Blank/whitespace briefing content: Feature-002 specifies that empty retry content is treated as generation failure, but current implementation/testing does not enforce this. Feature-003 must document the specified behaviour with a single expected-fail test and identify the mismatch as a defect. Do not fix the production behaviour as part of Feature-003.

Feature-001 SC-006: add verification of the existing non-generation request timing criterion using the approved mock repository/in-memory-store context.

Feature-001 SC-007: explicitly keep seam-substitution verification out of Feature-003 because the real US-12/US-14 implementation has not yet been substituted at that boundary. Record the deferral rather than silently leaving the criterion unresolved.

Explicitly keep out of scope:

Streamlit UI and advisor-facing presentation;

concurrent requests for the same student;

final Structured Advisor Briefing prompt content — US-12;

real Generative AI provider behaviour including latency, non-determinism, rate limits, filtering and token cost — US-13;

semantic briefing validation and real acceptance-criteria content — US-14;

live Unity Catalog Volume verification and full end-to-end component integration — US-16/infrastructure;

fixes to defects discovered by Feature-003.

Use repository findings to inform the specification, but do not turn low-level implementation details into specification requirements. Exact test filenames, fixture classes, helper implementations and harness structure belong in /speckit-plan.

If a repository behaviour conflicts with an already approved specification, preserve the specified behaviour and identify the implementation mismatch rather than rewriting the requirement to match the code.

If any genuine behavioural or acceptance ambiguity remains, leave it for /speckit-clarify rather than inventing an answer.

Do not modify implementation code or perform Git operations.

P03 - Clarify Feature-003 specification

Scope: Feature-003

Human review: Reviewed by Renny as part of the documented Feature-003 workflow.

Prompt Used (verbatim)

/speckit-clarify

Review the approved Feature-003 specification and identify only remaining material ambiguities that cannot already be resolved from:

the approved Feature-003 specification;

project constitution v1.1.0;

completed Feature-003 repository investigation;

Product Backlog US-18;

merged Feature-001 and Feature-002 artifacts;

my confirmed Feature-003 scope decisions.

Do not reopen decisions already explicitly resolved, including the testing-only scope, no production-code changes, use of controlled generation/validation test doubles, and deferred US-12/US-13/US-14/US-16 functionality.

Ask only questions requiring a genuine behavioural, testing, acceptance-criteria or workflow decision from me.

Do not ask about low-level test implementation choices that belong in /speckit-plan.

Do not invent answers. If no material ambiguity remains, state that clarification is complete rather than manufacturing additional questions.

Integrate my confirmed answers back into the specification.

P04 - Generate Feature-003 implementation plan

Scope: Feature-003

Human review: Reviewed by Renny as part of the documented Feature-003 workflow.

Prompt Used (verbatim)

Create the implementation plan for Feature-003: US-18: Structured Advisor Briefing Workflow Testing using the approved and clarified Feature-003 specification and project constitution v1.1.0.

Inspect the existing repository before determining the implementation structure. Use the completed Feature-003 repository investigation, merged Feature-001 and Feature-002 implementation, and confirmed human decisions as context.

Feature-003 is strictly a testing-only feature. It must not modify production source code or existing Feature-001/Feature-002 test files. Reuse the existing application architecture, test conventions and test doubles wherever possible, and determine the minimum new test/support files required.

Plan the implementation needed to verify the Structured Advisor Briefing workflow across generation, validation, retry, error handling and validated storage.

Clearly identify:

existing production and test files that remain read-only;

new Feature-003 test/support files that are genuinely required;

controlled generation and validation doubles used instead of US-13/US-14 implementations;

scenario coverage across service, REST and MCP boundaries without unnecessarily duplicating existing verified behaviour;

workflow call-order and retry-feedback verification;

storage write-count and valid-only persistence verification;

error mapping and metadata-only log-hygiene testing;

in-memory and Volume-backed store parity;

the Feature-001 SC-006 timing verification;

the blank/whitespace briefing expected-fail test and defect record;

the scenario-to-requirement/backlog traceability record.

Apply the approved clarification decisions:

the expected-fail annotation remains the authoritative record of the blank-content defect, with the traceability record also documenting it;

the traceability record may be hand-maintained but must include a verification that every referenced test/scenario name still resolves;

governed-storage parity testing should re-run the six approved storage-decision outcomes rather than every Feature-003 scenario.

Keep US-12 prompt content, US-13 real Generative AI behaviour, US-14 semantic validation, US-16 live end-to-end integration, Streamlit/UI testing, concurrency testing, SC-007 seam substitution and production defect fixes out of scope.

Treat implementation-level choices deliberately deferred from the specification as planning decisions. Do not introduce new production architecture, unnecessary abstractions or redundant test infrastructure.

If planning reveals that satisfying the approved specification would require modifying production code or existing Feature-001/Feature-002 tests, stop and identify the conflict for human review rather than expanding scope.

Do not implement tests or production code during planning.

Do not commit, push, merge, rebase, create pull requests or perform other Git/GitHub write operations.

P05 - Generate Feature-003 tasks

Scope: Feature-003

Human review: Reviewed by Renny as part of the documented Feature-003 workflow.

Prompt Used (verbatim)

Generate the actionable implementation tasks for Feature-003: US-18: Structured Advisor Briefing Workflow Testing from the approved and clarified Feature-003 specification and final implementation plan.

Treat the project constitution, spec.md, plan.md, supporting planning artifacts, repository findings and approved Feature-003 decisions as authoritative.

Feature-003 is strictly a testing-only feature. Generate only tasks required to implement the approved Feature-003 plan.

Requirements:

Preserve the exact scope, files, test architecture and implementation structure approved in the plan.

Use only the new Feature-003 test/support files identified by the plan.

Do not create tasks that modify production source code or existing Feature-001/Feature-002 test files.

Reuse existing test conventions, doubles and application seams rather than creating parallel testing infrastructure.

Order tasks by dependency and make them independently implementable and verifiable where practical.

Clearly identify tasks that can safely be completed in parallel.

Include tasks necessary to implement the approved scenario-driven workflow testing for:

successful first-attempt generation and storage;

validation failure followed by retry success;

terminal generation and validation failures;

configuration and storage failures;

not-at-risk and unknown-student behaviour;

get-or-create and explicit regeneration behaviour;

service, REST and MCP boundary verification where required by the plan.

Include the planned verification of:

workflow call ordering;

retry-feedback propagation;

storage write counts and prevention of invalid storage;

preservation of previously stored valid briefings;

error mapping and metadata-only log hygiene;

in-memory and Volume-backed store parity;

Feature-001 SC-006 timing;

the blank/whitespace expected-fail test and associated defect/traceability record;

traceability between scenarios, requirements, success criteria and backlog stories.

Preserve the approved clarification decisions:

the blank-content defect is recorded through both the expected-fail annotation and traceability record;

the hand-maintained traceability record includes verification that referenced test/scenario names resolve;

governed-storage parity re-runs only the six approved storage-decision outcomes.

Do not create tasks for US-12 prompt content, US-13 real Generative AI behaviour, US-14 semantic validation, US-16 live end-to-end integration, Streamlit/UI testing, concurrency testing, SC-007 seam substitution, or production defect fixes.

Do not duplicate behaviour already adequately verified by Feature-001 or Feature-002 unless the approved Feature-003 plan explicitly requires re-verification or cross-boundary evidence.

The resulting tasks must implement the approved Feature-003 plan, not reinterpret or expand it.

If task generation reveals a conflict between the approved specification, clarification decisions and plan, stop and identify it rather than silently resolving it.

Do not implement tests or code and do not perform Git/GitHub write operations.

P06 - Analyse Feature-003 before implementation

Scope: Feature-003

Human review: Reviewed by Renny as part of the documented Feature-003 workflow.

Prompt Used (verbatim)

Analyze Feature-003: US-18: Structured Advisor Briefing Workflow Testing for consistency across the approved constitution, specification, clarification decisions, final implementation plan, supporting planning artifacts, and generated tasks.

Feature-003 is strictly bounded to US-18 and remains a testing-only feature.

Check for:

requirements without corresponding tasks;

tasks without corresponding approved requirements;

contradictions between spec.md, plan.md, supporting planning artifacts and tasks.md;

inconsistencies with the approved Feature-003 clarification and planning decisions;

scope leakage into production-code changes, existing Feature-001/Feature-002 test modification, Streamlit/UI work, US-12, US-13, US-14, US-16, concurrency testing, SC-007, or defect fixes;

unnecessary duplication of behaviour already verified by Feature-001 or Feature-002;

missing coverage for the approved US-18 workflow scenarios, cross-boundary evidence, call ordering, retry-feedback propagation, storage decisions, error handling, log hygiene, store parity, SC-006, expected-fail defect recording and traceability;

inconsistencies in the handling of FR-012–FR-021 versus the no-duplication requirement, including whether the planned tracked-re-verification approach is supported by the approved specification;

missing acceptance-criteria or success-criterion coverage;

incorrect task dependencies or ordering;

unnecessary testing infrastructure or complexity;

constitution violations.

Confirm that all planned write targets remain limited to new Feature-003 test/support and traceability files, with production files and existing Feature-001/Feature-002 test files remaining read-only.

Verify that the blank/whitespace briefing mismatch remains an expected-fail test and documented defect rather than becoming a production fix.

Verify that controlled generation and validation test doubles remain the approved boundary and are not represented as real US-13 Generative AI or US-14 semantic validation.

If issues are found, identify the exact artifact, requirement/task and correction needed before implementation.

Do not modify implementation code.

Do not reinterpret or expand Feature-003 scope merely to resolve an inconsistency. If an approved artifact conflict requires a human decision, identify it explicitly.

Do not perform Git/GitHub write operations

P07 - Implement Feature-003

Scope: Feature-003

Human review: Reviewed by Renny as part of the documented Feature-003 workflow.

Prompt Used (verbatim)

Implement Feature-003: US-18: Structured Advisor Briefing Workflow Testing using the final approved constitution, specification, clarification decisions, implementation plan, supporting planning artifacts, analysis corrections, and tasks in specs/003-briefing-workflow-testing/.

Treat the current Feature-003 artifacts and the exact files identified by the approved plan/tasks as the implementation boundary.

Requirements:

Follow the approved specification, plan, contracts, planning decisions, analysis corrections, and tasks exactly.

Work through tasks in their approved dependency order.

Implement only the minimum testing functionality necessary to satisfy each approved task.

Modify or create only the new Feature-003 test/support and traceability files authorised by the plan/tasks.

Production source files and existing Feature-001/Feature-002 test files must remain unchanged.

Reuse the existing briefing generation, validation, retry, error-handling and storage architecture and existing testing conventions rather than creating parallel implementations.

Use controlled generation and validation outcomes as specified; do not implement US-13 real Generative AI generation or US-14 semantic validation.

Preserve the approved testing-only US-18 scope and do not implement Streamlit/UI testing, concurrency, SC-007, US-12, US-13, US-14, US-16, or unrelated functionality.

The previously recorded blank/whitespace briefing defect has already been resolved separately against the Feature-002 specification before Feature-003 implementation. Follow the latest Feature-003 artifacts: verify this behaviour as an ordinary passing test, retain the resolved defect in traceability, and do not treat it as an expected-fail test or reimplement the production fix.

Do not reinterpret, expand, or add requirements during implementation. Do not perform unrelated refactoring, cleanup, optimisation, dependency upgrades, architectural redesign, or speculative improvements.

If implementation appears to require modifying a file not authorised by the approved plan/tasks, stop before changing it and report the conflict for human review.

Complete all tests and traceability work required by the approved tasks, then run the specified pytest and Ruff validation gates.

After implementation:

report all files created or modified;

report completed task IDs;

report tests and validation commands run and their results;

report any unresolved or deferred items;

confirm that production source files and existing Feature-001/Feature-002 test files remained unchanged.

Then provide a human-review overview of Feature-003 so I can understand what was created and independently review the implementation.

The overview should:

summarise each new Feature-003 test/support and traceability file and its purpose;

explain how the tests verify generation, validation, retry, error handling and validated storage;

explain how controlled generation and validation test doubles work;

distinguish what Feature-003 verifies from functionality still deferred to US-13, US-14 and US-16;

explain how the resolved blank/whitespace briefing defect is now verified;

explain how service, REST and MCP behaviour is covered;

summarise how US-18, functional requirements, success criteria and test scenarios are traced together.

Finally, provide a practical manual verification guide for me to test Feature-003 myself.

Include:

the exact commands I should run;

which Feature-003 test files or test groups I should run;

what successful output should look like;

the most important individual scenarios I should inspect;

how I can manually confirm successful briefing workflow behaviour;

how I can confirm validation failure and retry behaviour;

how I can confirm terminal failure and error handling;

how I can confirm only valid briefings are stored and prior valid briefings are preserved;

how I can confirm in-memory and governed-store parity;

how I can confirm REST/MCP boundary behaviour and traceability;

how I can distinguish completed Feature-003 behaviour from deferred functionality.

Keep the manual verification guide concise and step-by-step so I can independently confirm that Feature-003: US-18 works according to the approved specification.

Do not commit, push, merge, rebase, switch branches, create pull requests, amend commits, or otherwise perform Git/GitHub write operations.

Appendix B - Manual Verification Guide

The following guide is preserved from the source Feature-003 worklog, with only the requested username replacement applied.

Manual Verification Guide (everything ran on powershell):

1. Run everything

```
cd "C:\Users\local_user\Projects\Capstone_T115\student_attrition_risk_app"
```

```
uv run ruff check .
```

```
uv run pytest
```

Expect: All checks passed! and 143 passed. No failures, no expected-fail.

2. Run Feature-003 alone

```
uv run pytest tests/test_briefing_workflow_outcomes.py tests/test_briefing_workflow_retry.py  
tests/test_briefing_workflow_storage.py tests/test_briefing_workflow_boundaries.py  
tests/test_briefing_workflow_traceability.py -v
```

Expect: 59 passed.

3. Confirm nothing forbidden was touched

```
git status --porcelain
```

Expect: exactly 7 ?? lines, no M lines.

```
git status --porcelain student_attrition_risk_app/src/
```

```
git status --porcelain student_attrition_risk_app/tests/ | Select-String -NotMatch '^?\?'
```

Expect: Return nothing

4. Successful workflow + ordering

```
uv run pytest tests/test_briefing_workflow_retry.py::test_first_attempt_success_engages_boundaries_in_specified_order -v
```

Read the assertion — it's the literal sequence store.has_validated → generate → validate → store.save_validated.

5. Validation failure → retry

```
uv run pytest tests/test_briefing_workflow_retry.py -k "feedback or propagated" -v
```

test_second_attempt_succeeds_only_when_feedback_propagated is the one to read: the pass depends on the marker, so it can't pass by accident.

6. Terminal failure and error handling

```
uv run pytest tests/test_briefing_workflow_outcomes.py -v
```

The 11 parametrised IDs are the full outcome matrix — scan them and check nothing's missing.

7. Storage decisions

```
uv run pytest tests/test_briefing_workflow_storage.py -k "write_count or survives" -v
```

8. Store parity

```
uv run pytest tests/test_briefing_workflow_storage.py -k governed -v
```

Expect: 6 passed — exactly the six approved outcomes, not more.

9. Boundaries and log hygiene

```
uv run pytest tests/test_briefing_workflow_boundaries.py -v
```

Note there are no REST tests — that's deliberate, and traceability.md § 2 says why.

Step 10 — prove the self-check works (PowerShell)

```
cd C:\Users\local_user\Projects\Capstone_T115\student_attrition_risk_app.
```

```
$t = "..\specs\003-briefing-workflow-testing\traceability.md"
```

Break a citation:

```
(Get-Content $t -Raw) -replace 'test_unknown_student_attempts_no_generation','test_this_does_not_exist' | Set-Content $t -Encoding utf8
```

Run the self-check — it should FAIL:

```
uv run pytest tests/test_briefing_workflow_traceability.py::test_every_cited_verification_resolves
```

Expect 1 failed, with the message naming test_this_does_not_exist as unresolved.

Undo:

```
(Get-Content $t -Raw) -replace 'test_this_does_not_exist','test_unknown_student_attempts_no_generation' | Set-Content $t -Encoding utf8
```

Confirm it passes again:

```
uv run pytest tests/test_briefing_workflow_traceability.py::test_every_cited_verification_resolves
```

Expect 1 passed.

⚠ Don't skip the undo — traceability.md is untracked, so git checkout can't restore it.

11. Offline confirmation

Disconnect your network and re-run uv run pytest. Expect identical results — 143 passed.

12. Completed vs deferred

Open specs/003-briefing-workflow-testing/traceability.md: § 1 is what Feature-003 verifies, § 2 is what was already covered, § 4 is what remains unverified and why.

Appendix B.1 - Evidence Scope

This condensed worklog is derived from the uploaded Feature-003 prompt-history document. It intentionally preserves the full development prompts in Appendix A, summarises long AI-generated outputs in the main worklog, preserves the manual verification guide, and reproduces the user-authored verification findings verbatim in Appendix C.

Appendix C - Human Verification Findings (verbatim)

Verbatim evidence note: The wording below is reproduced from the user-authored findings at the bottom of the source document. The only textual change is the requested replacement of the username with Renny.

Passed with expected output of 143 passes.

Passed with expected output of 59 passes.

Passed with expected exactly 7 ?? lines, and 1 M line, both verifiers to ensure no M line points at src/ with expected return nothing.

Passed with expected 1 passes.

Passed with expected test_second_attempt_succeeds_only_when_feedback_propagated test passed with 100%.

Passed with expected 11 parameterised IDs related to Functional requirements 12 to 21. (1. attempt-1-success, 2. retry-success, 3. terminal-generation, 4. terminal-validation, 5. configuration-failure, 6. storage-failure, 7. existing-returned, 8. regenerate-success, 9. regenerate-failure, 10. not-at-risk, 11. unknown-student).

Passed with expected output of 10 passes.

Passed with expected output of 6 passes.

Passed 7 tests, REST tests omitted in traceability.md.

Passed with 1 failed result and 1 passed after breaking it, inverted test.

Passed with expected output of 143 passes without network connection.

Verified traceability.md for Feature-003 §1 indicates completion, §2 is tracked re-verification, and §4 is what has been deferred.

All 12 manual verification checks have passed, completed by Renny for human overview. Therefore, can be committed and merged to GitHub repository origin/main branch.